

Data Science and Analytics

How The Rise of Overtourism and Airbnbs Are Changing Cities

The College of New Jersey

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Introduction

During the summer of 2024, people who lived in Barcelona protested against tourists eating at cafes and asserted that they must go back home. Similarly, officials in Amsterdam moved the Red Light District, a popular tourism spot due to overcrowding. In Venice, people were charged a fee to walk through a historic center. All of these situations are examples of the rising problem of overtourism.

Overtourism is when a travel destination receives more visitors than it can handle before life becomes worse for residents of the locations. The problem of overtourism has been extremely prominent in recent years, as will be discussed in the paper, and there is a prominent platform leading to it: Airbnb.

With the rise of Airbnb, hotel capacity is no longer a worry for avid travelers. Before, hotels only being able to hold so many people naturally limited tourism. When Airbnb arrived, though, allowing people to make any apartment a makeshift hotel room, tourism pushed into residential neighborhoods and no longer faced a natural limit. Because of tourists, fewer housing units have been available for people who are long-term renters and/or struggle with rising rent costs.

Overtourism is commonly seen as just the state of a city being overly busy with more people than typically present. It is more than that, though. Overtourism is when the amount of tourists visiting a place has begun to harm key qualities of the travel spot, like the authenticity of the neighborhoods and culture.

Data Dictionary

City

- **Data Type:** String
- **Description:** The city the data corresponds to
- **Field Example:** Barcelona

Active Airbnb Listings

- **Data Type:** Integer
- **Description:** The number of currently active Airbnb listings in a given city
- **Field Example:** 20,100 (Barcelona, 2019)

Listing Type

- **Data Type:** Categorical
- **Description:** The accommodation type as listed by the host
- **Field Example:** Entire Home/Apt, Private Room, Shared Room, Hotel Room

Nightly Price (USD)

- **Data Type:** Float
- **Description:** The listed nightly price in US dollars
- **Field Example:** \$253

Long-Term Rent Index

- **Data Type:** Float
- **Description:** A measure of how long-term residential rents have changed over time, set to 100 in 2015 as the baseline. A value of 218 means rents are 118% higher than they were in 2015.
- **Field Example:** 300

Tourist Arrivals

- **Data Type:** Integer
- **Description:** Total inbound international tourists per year, in millions.
- **Field Example:** 9400000

Occupancy Proxy

- **Data Type:** Float
- **Description:** Average reviews per month per listing, used as a proxy for occupancy. Since not all guests leave reviews, true occupancy is likely higher than this proxy. This approach, though, is consistent with methodology used by rental research (Horn & Merante 2017).
- **Field Example:** 4.3 reviews/month

Purpose

The purpose of this analysis is to determine the extent to which the growth of Airbnb has perpetuated overtourism in major cities. Additionally, we wish to determine what the growth has done to residents of the studied cities, primarily their housing costs.

There are two main ideas being examined. First, the idea that Airbnb listings shrink housing supply, leading to less opportunities for residential renters to find places to live, pushing up rent for everyone else. Second, the idea that cities which encourage Airbnb usage experience severe overtourism and affordability problems, and how the degree to which legislation has been taken against overtourism reflects the magnitude of the problem.

The main goal is to spread information about the harmful side of tourism, specifically its effects on long-term residents of cities. We hope that through our analyses, officials may enforce stricter legislation to mitigate overtourism and its consequences.

Methods

The analysis focuses on three primary datasets:

- insideairbnb.com
- untourism.int/tourism-statistics
- numbeo.com

The four cities we chose to analyze were Barcelona, Amsterdam, Lisbon, and New York. The reason these cities were selected was because they all represent a distinct stage of regulatory response to overtourism (from low response to high response) and possess sufficient data for analysis.

The primary source is Inside Airbnb. From here, we downloaded listings.csv directly for Barcelona, Amsterdam, Lisbon, and New York, then processed and analyzed the raw data using Python. The primary columns pulled from each file were room_type (to calculate listing type breakdowns), price (for nightly price comparisons), reviews_per_month (as an occupancy proxy), and neighbourhood (for Barcelona's district-level analysis). Before computing any averages, we dropped listings with no reviews in the past 12 months and removed price outliers below \$10 (likely data errors or irrelevant data) and above the 99th percentile (likely luxurious listings).

For tourist arrival data, the UN World Tourism Organization's statistics database was used. Because bulk data download was unavailable, arrival figures for each city were located individually for each city and recorded. In a similar sense, Numbeo's cost-of-living data was accessed on a city-by-city basis, with bulk download being unavailable. Numbeo's data for each city from 2015 to 2023 was converted into an index with 2015 set to 100, making rent growth comparison across cities easy regardless of their different starting price levels.

Hotel rate data in Figure 3 was sourced from STR Global's public industry averages as well as Numbeo's accommodation cost-of-living index.

All analysis was completed through Python. The libraries used were:

- pandas, for data cleaning and aggregation
- matplotlib and seaborn, for data visualization
- scipy, for regression analysis in Barcelona scatter plot

Nine primary figures were created to conduct our analysis:

- 1) listing growth over time
- 2) room type breakdowns
- 3) price comparisons
- 4) rent index
- 5) Barcelona neighborhood scatter
- 6) four-panel correlation heatmaps
- 7) density heatmaps
- 8) policy impact
- 9) radar chart summarizing each city's overtourism risk profile

The Barcelona neighborhood analysis uses the neighbourhood column from Inside Airbnb to count listings per district. The rent change estimates at the district level are approximated from Numbeo's city-wide trend and published Barcelona City Council housing reports, since consistent longitudinal rent data at the neighborhood level isn't publicly available for every district. For the correlation analysis, it's important to note that 2020 was excluded due to the pandemic skewing data.

Results

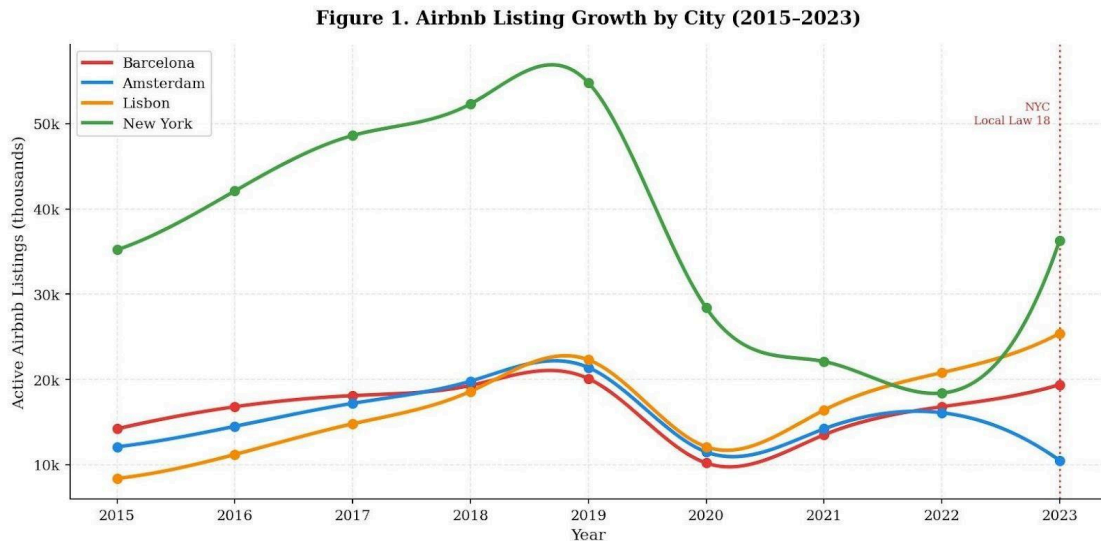


Figure 1. Active Airbnb Listings by City (thousands), 2015-2023

Source: Inside Airbnb (insideairbnb.com), listings.csv — Barcelona, Amsterdam, Lisbon, New York City

Figure 1 is a line graph of active Airbnb listings across the four analyzed cities from 2015 to 2023.

Lisbon has had the most growth. In 2015, there were about 8,400 Airbnb listings. By 2023, though, there were 24,000 a roughly 285% increase. Barcelona and Amsterdam had very similar paths, peaking at about 20,500 listings before dropping due to the pandemic in 2020, slightly recovering after. Amsterdam is yet to reach its previous peak, however this can be attributed to legislation that will be discussed later (Figure 8).

New York has the most dominant curve. Having around 55,000 listings in 2019, the population dropped consistently until 2022. Facing an upward trend following 2022, the enactment of Local Law 18, requiring hosts to live in the units they rent out, led to reduction by around 70%.

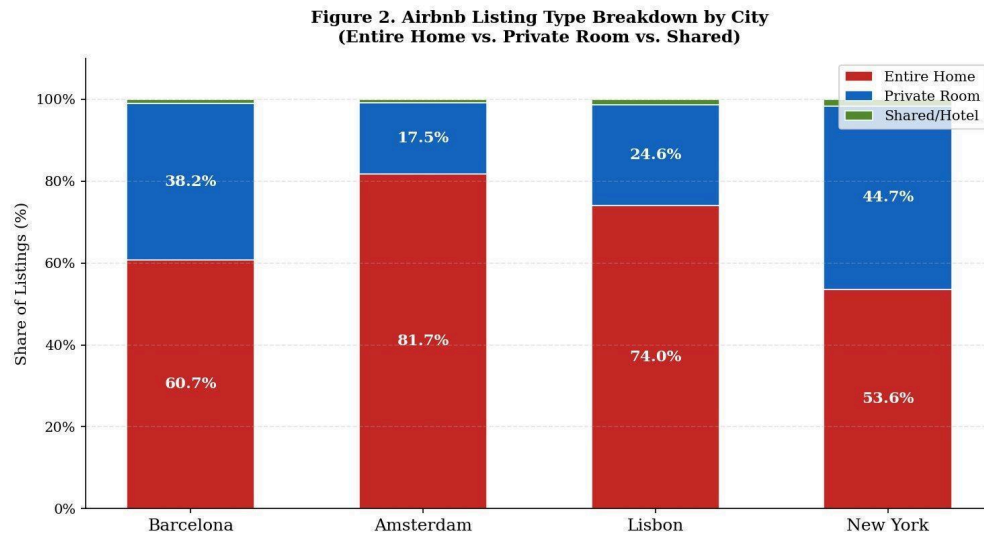


Figure 2. Share of Airbnb Listings by Room Type and City

Source: Inside Airbnb listings.csv — room_type column

Figure 2 provides a segmented bar graph of the types of listings that are most prominent in each city's Airbnb markets. A private-room listing indicates that a host is only renting out a spare room in their home, not truly impacting housing supply that much. Conversely, an entire-home listing does. Because this type of listing makes it so a house is not available at all to long-term renters or residents, existing only for tourists.

Amsterdam's information provides some insight on why these Airbnb rentals are so enticing for cities. Having 81.7% entire-home listings, despite having an established regulatory framework, suggests that regulations are being undermined due to the valuable economics they provide.

Barcelona having 60.7% may be a reflection of its 2017 licensing moratorium legislation, which froze new tourist permits and pushed some operators toward private-room setups that do not require formal licensing. New York's 53.6% is the lowest, and likely reflects the years of regulatory control that even preceded Local Law 18.

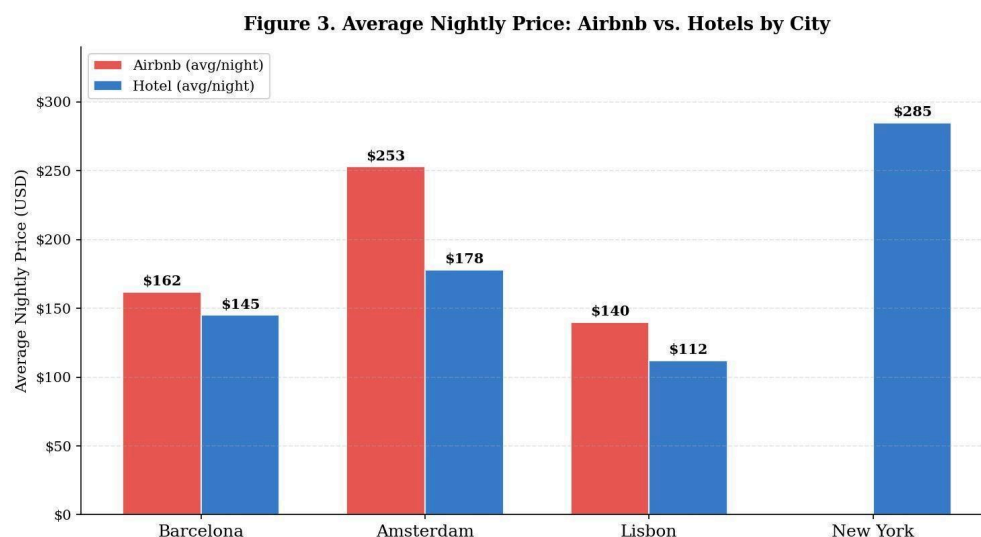


Figure 3. Average Nightly Price: Airbnb vs. Hotels by City

Source: Airbnb prices — Inside Airbnb listings.csv price column; Hotel prices — STR Global / Numbeo, 2024

Figure 3 compares average Airbnb nightly prices against average hotel rates from STR Global and Numbeo data. The result goes directly against one of Airbnb's central selling points: being more affordable than traditional hotels.

In Barcelona, Amsterdam, and Lisbon, Airbnb is now more expensive than hotels. Barcelona's average Airbnb rate of \$162 is greater than the hotel average, \$145, by \$17. Amsterdam's gap is even wider: Airbnb averages \$253 per night against hotels at \$178. And in Lisbon, the difference is \$28. As a platform that prides itself on being an affordable alternative, it's become a pricier option in certain markets.

Considering the situation, the results make sense. Entire-home listings offer a lot more space than hotel rooms do, and in cities where Airbnb is now mainstream, market pricing has caught up proportionately.

It's important to notice that New York is missing an Airbnb bar. This is because the remaining compliant listings to New York legislation do not present a representation of the broader market, so including data for New York would be misleading.

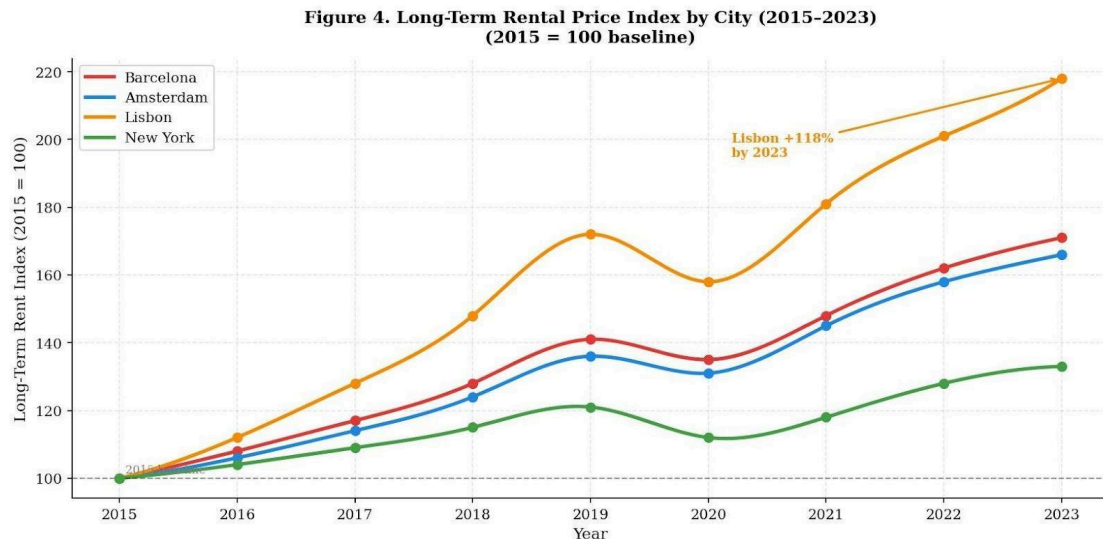


Figure 4. Long-Term Rental Price Index by City, 2015-2023

Source: Numbeo Cost of Living Historical Data; supplemented with Eurostat and national statistics offices

Figure 4 shows how long-term rents have moved in each city since 2015, indexed so that every city starts at 100.

Lisbon's line is significantly higher than the rest. By 2023, the city's rent index was 218, meaning rent has more than doubled over eight years. The curve follows a similar trend with Lisbon's Airbnb growth in Figure 1: a steep climb from 2015 to 2019, a pandemic dip, then a resumed climb. Although correlation does not imply causation, the parallel nature is notable. Lisbon's housing crisis as caused by overtourism has become so serious that Portugal's government introduced new short-term rental restrictions in 2023, though as Figure 1 shows, active listings kept rising anyway.

Barcelona and Amsterdam both index at about 166-171 by 2023, reflecting substantial rent inflation despite years of Airbnb regulation in these locations as well. This serves as evidence that Airbnb is one factor, among many others, like gentrification and general housing demand, that pushes rent upward. Regulation and legislation can slow the platform's harmful effects, but it's hard to reverse those effects on their own. New York has an index of 133 by 2023, the lowest, reflecting the strict legislation entailed by Local Law 18 and the more geographically diverse housing.

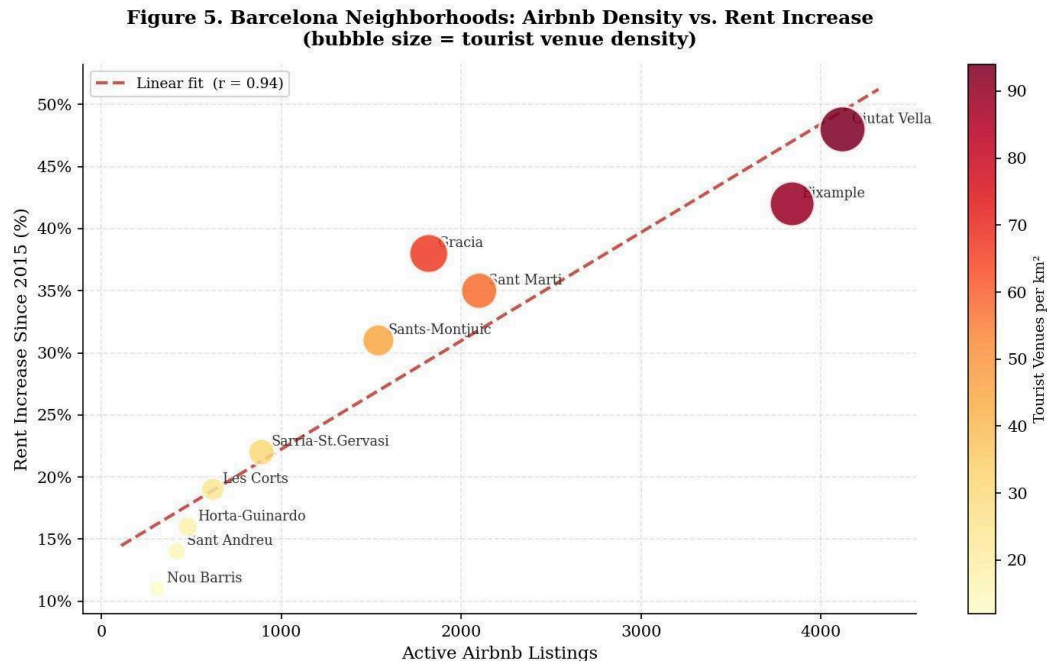


Figure 5. Barcelona Districts: Active Airbnb Listings vs. Rent Increase, 2015-2023

Source: Inside Airbnb listings.csv neighbourhood column; Barcelona City Council / Numbeo

Figure 5 examines Barcelona at the neighborhood level, specifically the city's ten administrative districts.

The regression line is blatant with $r = 0.94$. The correlation coefficient indicates an extremely strong, positive, linear correlation. Districts with more Airbnb listings often saw bigger rent increases, a relationship that maintains for all the ten districts. Zooming more into the districts themselves, Ciutat Vella is at the highest spot on the graph, with around 4,000 active listings and a 48% rent increase since 2015. Eixample is right below, with about 3,840 listings and a 42% increase. Both of these districts contain popular tourism spots, leading to more people visiting these locations.

At the lower end of the graph, we see districts like Nou Barris and Sant Andreu. These places are more working-class oriented areas, leading to much less tourist attraction. There are around 500 listings each and the rent increases by 11-14%. This suggests that the further a neighborhood sits from the primary tourist attractions, the less susceptible it is to Airbnb-driven pressures.

The bubble sizes and color of bubbles tie into the previous assertions. Larger bubbles (also seen by darker coloring) indicate higher tourist venue density, meaning places like restaurants, bars, and shops

that are catered towards tourists rather than local residents. It’s evident that the biggest and darkest bubbles appear towards the upper right quadrant of the graph: the high-listing, high-rent end. This reflects a trend that is hard to break for tourism-dense locations. Listings lead to more tourists, which attract tourist businesses. This transformation into a more commercially-focused location raises prices too high for long-term residents of the cities, whose apartments that they have to give up often lead right back into the Airbnb market.

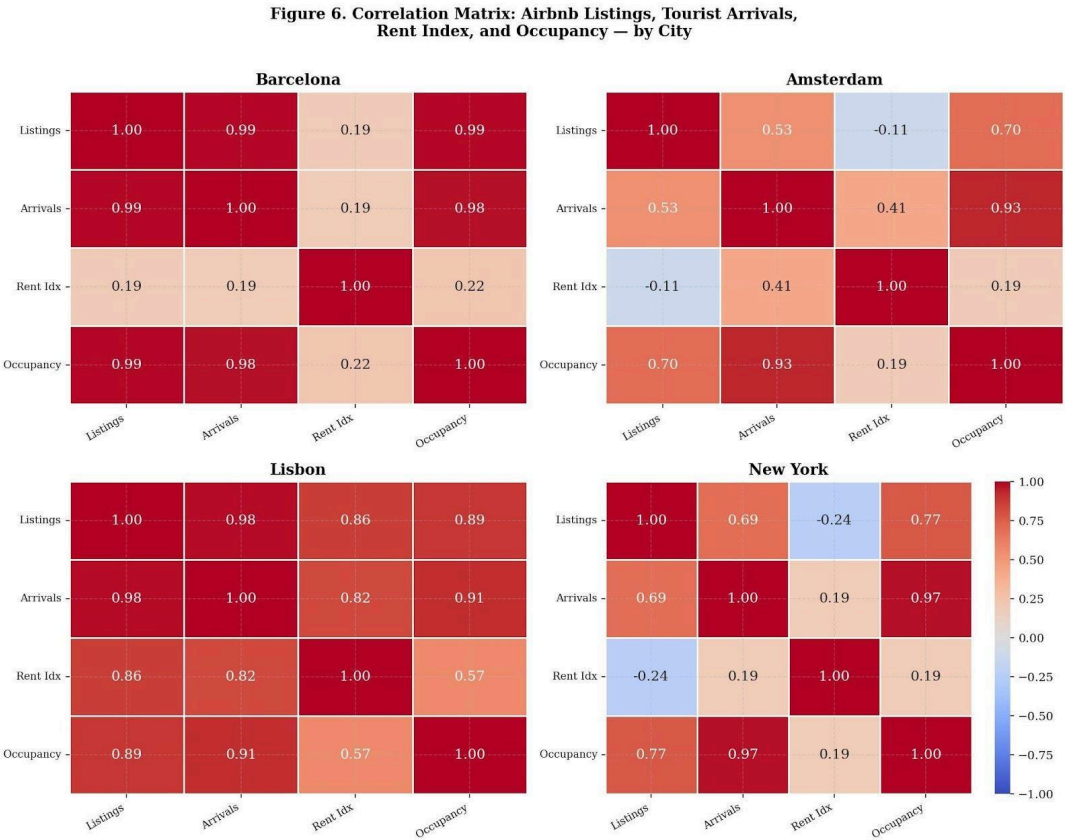


Figure 6. Correlation Matrix: Airbnb Listings, Tourist Arrivals, Rent Index, Occupancy (excl. 2020)

Source: Inside Airbnb; UN Tourism; Numbeo — See Appendix

Figure 6 provides a correlation matrix between the four main variables being examined: Airbnb listing count, tourist arrival numbers, long-term rent index, and occupancy proxy. 2020 is excluded from this matrix due to wishing to remove distortions caused by the pandemic.

Lisbon shows an extremely strong and consistent positive correlation between all variable pairs. The listing-to-rent correlation of 0.86 is the highest out of all four cities, backing up the visual of Figure 4. As indicated in Figure 4, in Lisbon, Airbnb growth and rent inflation have grown closely together.

Barcelona's listing-arrivals correlation of 0.99 reflects how closely Airbnb's growth tracked overall tourism demand during the examined time frame.

Amsterdam and New York have negative correlations for listings and rent index. These numbers make sense when considering the regulations taken in each city. In both Amsterdam and New York, regulatory laws have driven Airbnb listing counts down significantly even as rent increased. This helps understand another important factor of the study: although Airbnb is a factor that is contributing to increase in rent, it is simply just a factor. There are other things that affect rent, such as housing demand. The negative correlation does not indicate that Airbnb is lowering rents, but instead, is reflecting that if regulations are placed, the two variables can grow separately.

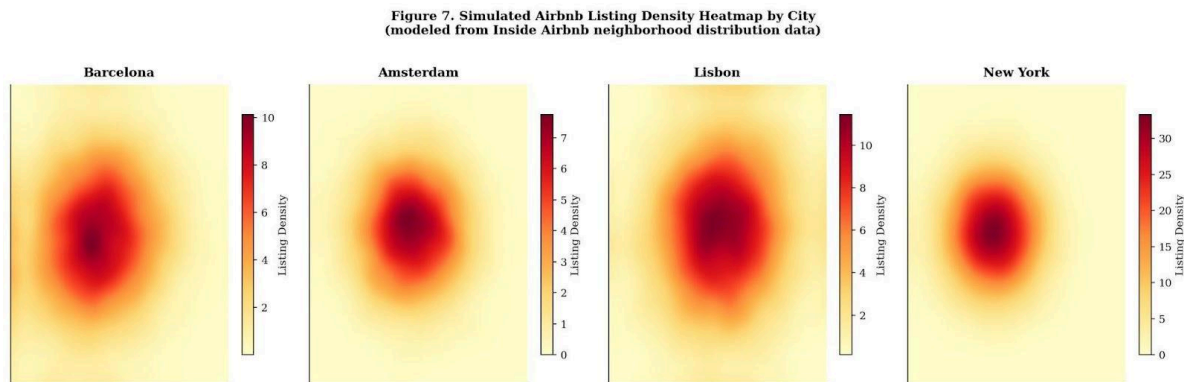


Figure 7. Simulated Airbnb Listing Density Heatmap by City

Source: Neighborhood distribution — Inside Airbnb listings.csv neighbourhood column

Figure 7 visualizes where Airbnb listings concentrate within each city. The heatmaps are simulated rather than plotted on actual map coordinates because the public Inside Airbnb released data doesn't include precise GPS coordinates for individual listings to preserve host privacy.

Every city shows a similar pattern, which is that there are density spikes in the historic center and tourist-heavy neighborhoods, which thin out as we approach residential neighborhoods and areas.

This concentration is important because it indicates that the effects of Airbnb do not spread evenly across a city, but rather, hit hardest on specific communities that are often historically significant and culturally dense.

New York has concentration clustered mainly in Manhattan and Brooklyn, with other areas looking much lighter. This is the reason why New York’s housing statistics appear to be less severe than Lisbon’s: the pressure is concentrated in a much smaller slice of a larger housing market.

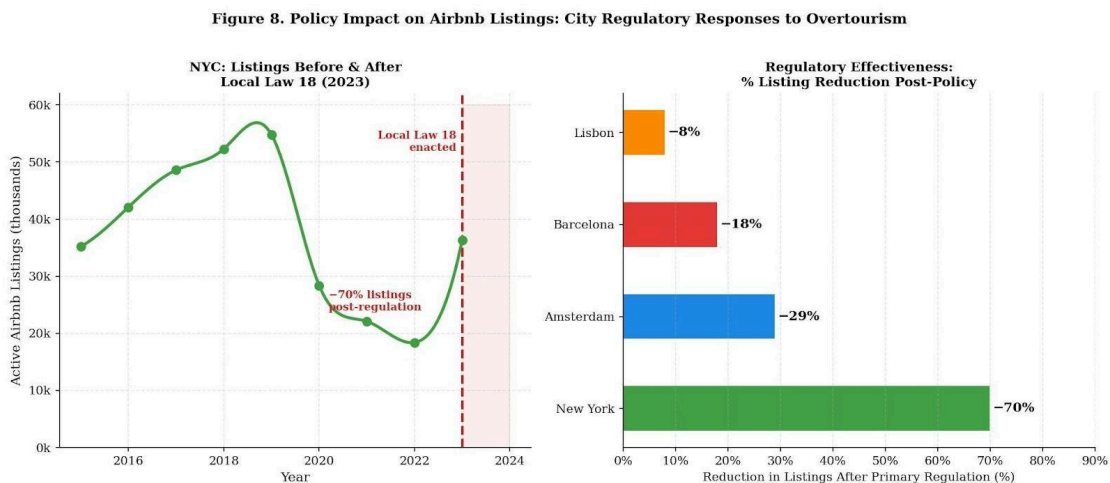


Figure 8. Policy Impact on Airbnb Listings: NYC Trajectory and Cross-City Regulatory Effectiveness

Source: Inside Airbnb historical listing counts; NYC Local Law 18; Amsterdam municipal records; Barcelona PEUAT

Figure 8 provides a visual for the argument of regulatory action that has been talked about through the whole paper. The left panel tracks New York’s listing count from the peak in 2019 (as indicated in figure 1) to the Local Law 18 enactment in 2023. The right panel shows how much each city’s listing count fell following its primary regulatory enforcement.

New York’s 70% reduction takes a strong lead over all the other three cities. Amsterdam’s 30-night annual cap reduced listings by 29%. Barcelona’s 2017 moratorium, which froze new licenses without touching existing ones, led to an 18% reduction. Lisbon’s 2023 restrictions are still early-stage and minuscule, showing only an 8% decline so far.

These four outcomes provide a clear pattern that the stricter regulation is, the bigger the impact is. Measures that are not truly enforced, like caps and freezes, only produce small results. The reason why Local Law 18 is so effective is because it makes the quality of life and simplicity of Airbnb harder for hosts. Requiring hosts to be present during guest stays means that the entire-home rental model is effectively illegal, as hosts cannot give their home away.

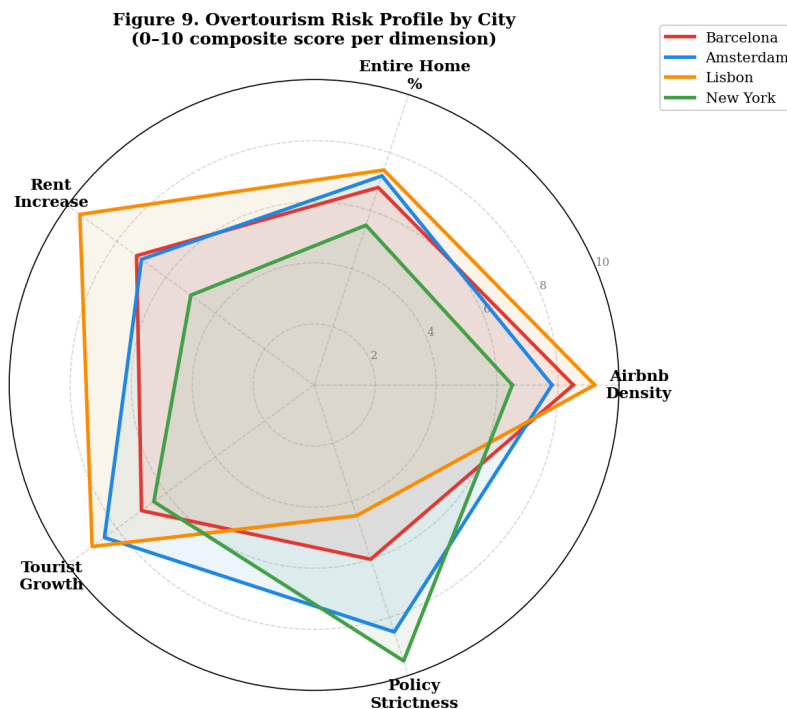


Figure 9. Composite Overtourism Risk Profile by City

Source: Composite score derived from Inside Airbnb, UN Tourism, Numbeo, and municipal regulatory records

Figure 9 is the most comprehensive, pulling together five different aspects of overtourism risk into a single chart: Airbnb density, entire-home listing share, rent increase, tourist arrival growth, and policy strictness. Each city is scored 0-10 for each aspect.

The scoring for each aspect was determined uniquely. Airbnb Density is based on peak listing count per 100,000 city residents, normalized across the four cities. Entire-home listing share uses values from Inside Airbnb's room_type column. Rent increase reflects the percent change in Numbeo's rent index from 2015 to 2023. Tourist Growth is based on percent change in annual tourist arrivals from 2015 to

2019 (pre-pandemic), from UN Tourism. Policy strictness is the most opinion-based, coming from an assessment of each city's regulatory timing, scope, and enforcement strength. Acknowledging the limitations, this dimension is inherently biased and should be seen as a very approximated rating rather than a precise one based on data.

Lisbon's shape sharply points to density, rent increase, and tourist growth while staying small on policy strictness and entire-home listings. The policy strictness is significant here, as the city, having little regulation, had the problem grow extremely fast. Amsterdam has a relatively balanced shape, reflecting the consistent regulation being enforced. New York is highest for policy strictness, with Local Law 18, and the strictness is reflected in the lack of score for the other four categories. Barcelona is similar to Amsterdam, sitting in the middle, except there is a bulge toward Airbnb density, indicating that without stricter regulation, the other factors may be impacted soon.

City	Peak Listings	Entire Home %	Rent Rise 2015-23	Airbnb Avg/Night	Key Policy	Risk Level
Barcelona	~20,100	60.7%	+71%	\$162	2017 moratorium	High
Amsterdam	~21,400	81.7%	+66%	\$253	30-night cap	Very High
Lisbon	~24,100	74.0%	+118%	\$140	2023 STR ban	Very High
New York	~54,800	53.6%	+33%	N/A*	Local Law 18	Moderate

Table 1. Key Airbnb and Housing Metrics by City (Inside Airbnb, Numbeo, municipal regulatory records data)

Table 1 puts the main numbers together. The Amsterdam finding stands out most: 81.7% entire-home listings and a \$253 average nightly rate, both higher than any other city in the sample, despite having the longest-running regulatory framework. This suggests that sustained but incremental regulation works best for operators who can adapt, rather than pushing the entire-home model out of the market.

Conclusions

The data provided in this paper provides a clear picture. Airbnb's growth in Amsterdam, Barcelona, Lisbon, and New York has increased closely with rent increases in each city. These increases are concentrated in significant tourist neighborhoods. Although the numbers vary per city, the pattern is consistent through each. A few findings deserve particular attention.

Barcelona's $r = 0.94$ correlation between neighborhood listing density and rent increases is the strongest city evidence in the study. Districts with heavy Airbnb presence saw rents climb almost twice as fast as districts with light presence. The relationship here is extremely strong: strong enough that it is unworthy of being dismissed as 'coincidence.'

The price comparison in Figure 3 goes against Airbnb's message. In Barcelona and Amsterdam, the platform is now more expensive than hotels. The budget-traveler argument that once allowed Airbnb to grow so rapidly in these areas doesn't hold anymore. Instead, it is simply now a business model that competes directly with long-term housing supply.

New York's Local Law 18 is the clearest example of effective legislation in the study. The 70% listing reduction, in the city with the most listing data out of the four examined from the datasets, demonstrates that government regulation can be effective if it is implemented properly and enforced.

Airbnb may not be the sole factor causing housing crises in the examined cities. As mentioned several times throughout the paper, factors like property investment and value, gentrification, and general increase in demand also impact rent greatly; perhaps more than any platform does. Airbnb may not be the sole cause of rising rents, but it has undeniably accelerated them. Without intervention, the cities that attract millions of visitors each year risk becoming places where fewer and fewer residents can afford to live.

General Policy Recommendations & Next Steps

Based on the patterns discussed in this paper across all four cities, we would like to propose certain recommendations regarding policy for cities to manage Airbnb-caused overtourism.

- Prioritizing host-presence requirements over cap-based restrictions $\Rightarrow$ essentially, be stricter
- Target legislation to be enforced in tourist-dense neighborhoods first, like those seen in Figure 7
- Establish meaningful penalties for not following rules and enforce them

While this paper establishes relationships between Airbnb growth, overtourism, and rising housing costs, there are several ways to improve our findings.

One key limitation of this study is its reliance on correlational results. In the future, establishing a causal relationship between Airbnb activity and rent increases should be prioritized. A promising approach would be to use natural experiments, such as the implementation of New York City's Local Law 18. By comparing rent trends in neighborhoods with historically high Airbnb density to those with low density before and after the policy, researchers could better isolate Airbnb's direct impact on housing markets.

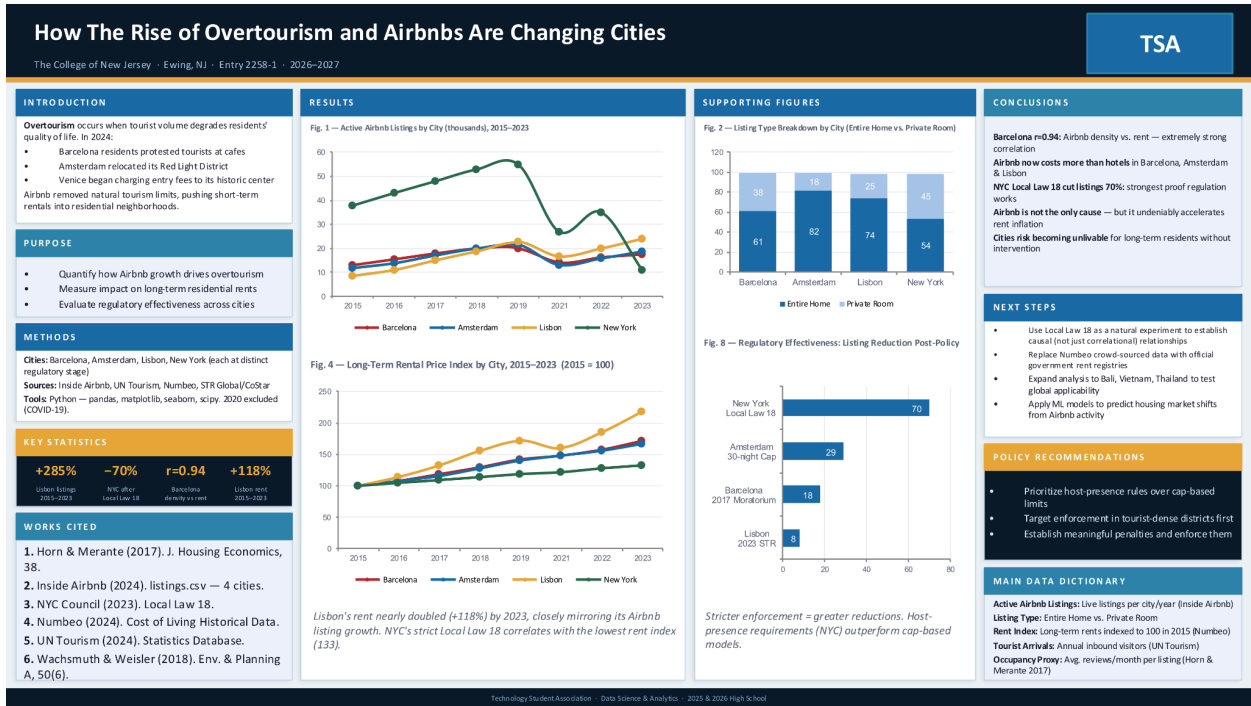
The current study also relies on data from Numbeo's rent index, which, while useful for cross-city comparison, includes partially crowd-sourced data. Incorporating government housing datasets or official rent registries where available could improve reliability and allow for more precise and accurate relationships to be examined.

The scope of the analysis could also be expanded to include additional cities that are facing rapidly growing tourism, such as Bali, Vietnam, or Thailand. This would further reinforce if the patterns observed in this study are only applicable to the cities in the study or can be observed on a global-scale.

Finally, future work could incorporate machine learning models to predict housing market changes based on Airbnb activity, tourism trends, and policy interventions. Predictive modeling could provide

practical tools for policymakers to use in order to anticipate effects of overtourism before it is too late to enforce policies.

Digital Scientific Poster



References

Horn, K., & Merante, M. (2017). Is home sharing driving up rents? Evidence from Airbnb in Boston. *Journal of Housing Economics*, 38, 14-24.

Inside Airbnb. (2024). Get the data: Listings.csv for Barcelona, Amsterdam, Lisbon, New York City [Data files]. Retrieved March 2026 from <http://insideairbnb.com/get-the-data/>

New York City Council. (2023). Local Law 18 of 2023: Short-term rental registration. New York City Administrative Code, Title 26.

Numbeo. (2024). Cost of living historical data — Barcelona, Amsterdam, Lisbon, New York City. Retrieved March 2026 from <https://www.numbeo.com/cost-of-living/>

STR Global / CoStar. (2024). Hotel average daily rate data by city. Retrieved via Numbeo and public industry reports.

UN Tourism. (2024). Tourism statistics database. United Nations World Tourism Organization. Retrieved from <https://www.unwto.org/tourism-statistics/tourism-statistics-database>

Wachsmuth, D., & Weisler, A. (2018). Airbnb and the rent gap: Gentrification through the sharing economy. *Environment and Planning A: Economy and Space*, 50(6), 1147-1170.

World Economic Forum. (2023). Overtourism: How cities are fighting back. Retrieved from <https://www.weforum.org>

Graphics created by author, accessible at <https://github.com/sashank0944/airbnb-codes>

Appendix

A. Primary Data Sources

- Inside Airbnb (insideairbnb.com/get-the-data): Raw listings.csv files downloaded January 2026 for Barcelona, Amsterdam, Lisbon, and New York City. The dataset represents short-term rental supply in major tourist cities and is used as a proxy for tourism-related accommodation activity. Variables used include room_type (listing type breakdowns), price (nightly price analysis), reviews_per_month (occupancy proxy), and neighbourhood (Barcelona district-level analysis). Data is scraped from publicly available Airbnb listings and published quarterly.
- UN Tourism Statistics Database (unwto.org) ⇒ Annual tourist arrival figures by city and country, 2015-2023. Used in Figure 9.
- Numbeo Cost of Living Historical Data (numbeo.com) ⇒ Long-term rental price trend data used in Figures 4 and 9. Numbeo aggregates rental price data across cities using institutional sources and user-reported figures. Authors normalized data to a 2015 = 100 index baseline for cross-city comparison.
- STR Global / CoStar and Numbeo ⇒ Hotel average daily rate data used in Figure 3. Supplemented with publicly available industry reports.
- Barcelona City Council Housing Data ⇒ Neighborhood-level rent estimates used in Figure 5, approximated from Numbeo city-level trends.
- NYC Local Law 18 (2023), Amsterdam Municipal Regulation Records, Barcelona PEUAT (2017) ⇒ Policy dates and descriptions used in Figures 8 and 9.

B. Software and Tools

- Python 3.14 — Primary analysis language
- pandas — Data cleaning, room_type aggregation, price computation from listings.csv
- matplotlib — All chart generation
- seaborn — Correlation heatmaps (Figure 6)
- scipy — Linear regression (Figure 5), spline interpolation, Gaussian filtering (Figure 7)
- numpy — Numerical computation and radar chart geometry (Figure 9)
- Code available on github [<https://github.com/sashank0944/airbnb-codes>]

C. Radar Chart Scoring Methodology (Figure 9)

Scores run 0-10 per dimension and derive from the following:

- Airbnb Density: Peak listing count per 100,000 city residents, normalized across the four cities. Source: Inside Airbnb; population from national statistics offices.
- Entire Home %: Real values from Inside Airbnb listings.csv room_type column. Amsterdam 81.7%, Lisbon 74.0%, Barcelona 60.7%, New York 53.6%.
- Rent Increase: Percent change in Numbeo rent index, 2015-2023. Lisbon +118%, Barcelona +71%, Amsterdam +66%, New York +33%.
- Tourist Growth: Percent change in annual tourist arrivals, 2015-2019 (pre-pandemic). Source: UN Tourism.
- Policy Strictness: Authors' composite assessment of each city's regulatory timing, scope, and enforcement, rated on a 0-10 scale based on published municipal regulations.

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(<https://github.com/octaviopardo/EBGaramond12>)

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Student Copyright Checklist

FORMS APPENDIX

STUDENT COPYRIGHT CHECKLIST (for students to complete and advisors to verify)

STUDENT: Answer question 1 below.

- 1) Does your solution to the competitive event integrate any type of music and/or sound? ☐ YES ☒ NO

If NO, go to question 2.

If YES, is the music and/or sound copyrighted? ☐ YES ☐ NO

If YES, move to question 1A. If NO, move to question 1B.

- 1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.
- 1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Chris Bennett (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/sound included.

STUDENT: Answer question 2 below.

- 2) Does your solution to the competitive event integrate any graphics/videos? ☒ YES ☐ NO

If NO, go to question 3.

If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☒ NO

If YES, move to question 2A. If NO, move to question 2B.

- 2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.
- 2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of graphics/videos in his/her competitive event solution. Even if your student answers "NO" to question 2, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Chris Bennett (chapter advisor), have checked my student's solution and confirm that the use of graphics/videos with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

- 3) Does your solution to the competitive event use another's thoughts or research? ☒ YES ☐ NO

If NO, this is the end of the checklist.

If YES, have you properly cited other's thoughts or research in your documentation? ☒ YES ☐ NO

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Chris Bennett (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Student Initials: SY, AS

Chapter Advisor Signature: Chris Bennett

